Biology and function of the aryl hydrocarbon receptor: report of an international and interdisciplinary conference.

The aryl hydrocarbon receptor (AhR) is a ligand-activated transcription factor present in many cells. The AhR links environmental chemical stimuli with adaptive responses, such as detoxification, cellular homoeostasis or immune responses. Furthermore, novel roles of AhR in physiological and genetic functions are being discovered. This is a report of a recent meeting in Düsseldorf. The meeting highlighted that AhR research has moved from its focus on toxic effects of dioxins and other environmental pollutants to its biological roles. For instance, it was recently discovered that AhR-responsive elements in retrotransposons contribute to the functional structure of the genome. Other exciting new reports concerned the way plant-derived compounds in our diet are necessary for a fully functioning immune system of the gut. Also, human brain tumours use the AhR system to gain growth advantages. Other aspects covered were neurotoxicology, the circadian rhythm, or the breadth of the adaptive and innate immune system (hematopoietic stem cells, dendritic cells, T cells, mast cells). Finally, the meeting dealt with the discovery of new xenobiotic and natural ligands and their use in translational medicine, or cancer biology and AhR.